

VSync would be enabled only at a certain performance range.

When a GPU and a monitor are working together there can be three situations:

- ◆ The GPU refresh rate is lower than the minimum refresh rate of the monitor.
- ◆ The GPU refresh rate is between the minimum and the maximum refresh rate of the monitor.
- ◆ The GPU refresh rate is higher than the maximum refresh rate of the monitor.

Situation 2 allows the solution of the GPU setting the refresh rate of the monitor to match its own. In case 3, the GPU relies on the standard VSync mode and lowers its refresh rate to be within range

of the monitor. For the first situation though, NVIDIA puts in an interesting fix by controlling the refresh rate of the monitor to be an integral multiple of that of the GPU. This is sort of like the VSync solution turned on its head.

All this capability comes at a price (quite literally). NVIDIA needs to have a proprietary chip in every monitor that intends to use its technology. Needless to say, this drives up the price of the displays thanks to the licensing fees they have to shell out to NVIDIA. Also, AMD GPU's don't support this. AMD had to strike back with this and their very own solution to the Sync problem.



WHAT PURPOSE DOES THE GAMMA SETTING SERVE?

Originally incorporated in the source material itself to compensate for the non-linear brightness response of CRT televisions, gamma correction isn't required in modern flat panel TVs which are capable of perfectly linear brightness levels. Because the overwhelming majority of source material did include gamma correction, modern flat panel displays have to carry this feature for the purpose of compatibility. The ideal gamma level is 2.2, setting it lower causes the image to wash out, whereas increasing it leads to punchier high-contrast picture, albeit with reduced black and white details.

FreeSync - AMD's framed solution

Not only did AMD throw in their gloves by ensuring incompatibility by NVIDIA's G-Sync, they also struck back hard by coming up with their own alternative FreeSync, which was based on AdaptiveSync, part of open standard DisplayPort™ 1.2a video interface from Video Electronics Standards Association (VESA®). AMD specifically collaborated with VESA to